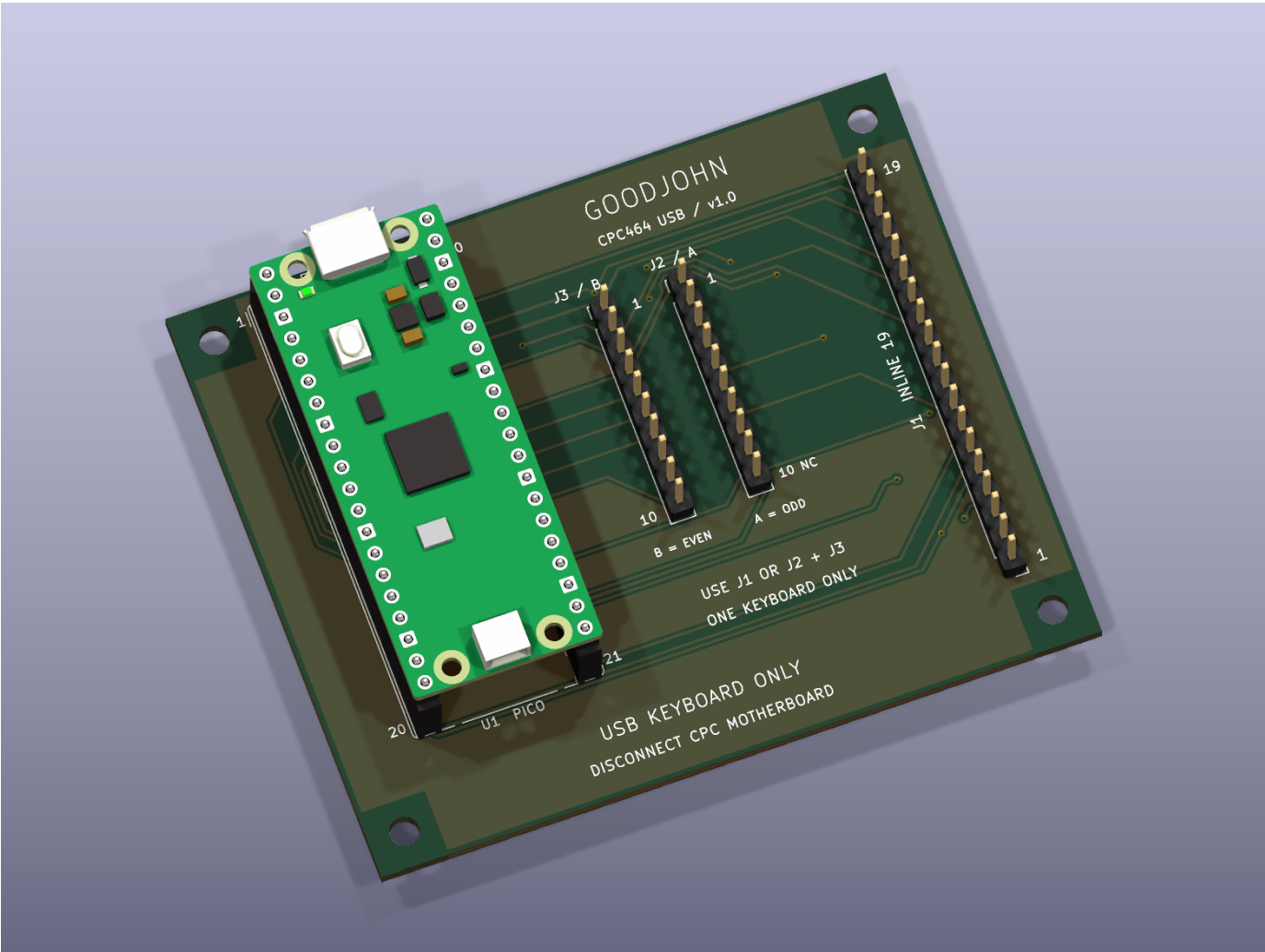


GOODJOHN / PICO CARRIER v1

Assembly guide - 2026-10-06 - UNBUILT PROTOTYPE

USB KEYBOARD ONLY: disconnect the CPC motherboard, even when off.
Use one keyboard via J1 OR J2 + J3. No CPC pass-through.



KiCad render; 80 x 66 mm board. Four 3.2 mm mounting holes.

Fit to carrier	Quantity / specification
U1 module	1 x original RP2040 Pico / Pico H
U1 female sockets	2 x 1-by-20; 2.54mm pitch; 17.78mm row spacing
J1 male header	1 x 1-by-19, 2.54mm; pin 1 at BOTTOM
J2 / A and J3 / B	2 x 1-by-10, 2.54mm; pin 1 at TOP
USB	One PC-to-Pico USB data cable supplies power

Pico USB faces the top edge. Insert U1 only after checking the unpowered carrier.
Carrier headers take a mating harness, not a bare membrane tail.
Hole centres from top-left: (4,4), (76,4), (4,62), (76,62) mm.
Check actual sockets/cable fit; the carrier has not been bench-tested.

CONNECTOR CHECK SHEET

All numbers below are PHYSICAL connector contacts. GPIO is a separate column.

Net	GPIO	Pico pin	J1 inline	Ribbon carrier	Bread80 J1
X9	GP20	26	1	J2.9	17
Y10	GP11	15	2	J2.1	1
X1	GP12	16	3	J2.2	3
X2	GP13	17	4	J2.3	5
X3	GP14	19	5	J2.4	7
X4	GP15	20	6	J2.5	9
X5	GP16	21	7	J2.6	11
X6	GP17	22	8	J2.7	13
X7	GP18	24	9	J2.8	15
X8	GP19	25	10	J3.1	2
Y9	GP10	14	11	J3.2	4
Y8	GP9	12	12	J3.3	6
Y7	GP8	11	13	J3.4	8
Y6	GP7	10	14	J3.5	10
Y5	GP6	9	15	J3.6	12
Y4	GP5	7	16	J3.7	14
Y3	GP4	6	17	J3.8	16
Y2	GP3	5	18	J3.9	18
Y1	GP2	4	19	J3.10	20

J2.10 = A10 = Bread80 J1 pad 19: NO CONNECTION.

J3.10 = B10 = Bread80 J1 pad 20: Y1, required.

Bread80 ribbon mode: LK1 1-2, LK2 1-2; LK3 and bonus links OPEN.

Before USB: remove Pico, disconnect CPC, then continuity-check every cable path.

Different matrix nets must be isolated; none may connect to GND or power.

Attach keyboard only after those checks. Key continuity appears only when held.

Diode keyboards may need diode-test mode and the correct probe polarity.

Key	Inline carrier contacts	Pico physical pins
1	J1.3 + J1.11	16 + 14
2	J1.4 + J1.11	17 + 14
Esc	J1.5 + J1.11	19 + 14
Main Enter	J1.5 + J1.17	19 + 6
Space	J1.10 + J1.14	25 + 10
DEL / Backspace	J1.1 + J1.2	26 + 15

After checks, fit Pico and use the existing Goodjohn firmware with ghost suppression.

ERC / DRC / pin-map checks passed; physical fit and keyboard tests remain pending.